SMART INDIA HACKATHON 2025

AI Metro Induction System

- **Problem statement ID-** SIH25081
- **Problem Statement Title-** AI-Driven Train Induction Planning & Scheduling for Kochi Metro Rail Limited (KMRL)
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team Name-** CODEZEN

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AI Metro Induction System

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❖ Proposed Solution :

- Our solution is a React-based AI decision support system that automates train induction planning for Kochi Metro.
- It evaluates each train's readiness using multiple real-time operational parameters such as:
- Fitness Certificates (Rolling Stock, Signalling, Telecom validity)
- Job Card Status from maintenance logs

❖How It Addresses the Problem :

- Eliminates manual decision errors in train readiness evaluation.
- Provides real-time alerts on expiring certificates or critical issues.
- Data export feature for audit and record-keeping.
- Improves operational efficiency by automatically sorting trains by priority.

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TECHNICAL APPROACH

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- **Language**: JavaScript (ES6)
- **Framework**: React.js (for dynamic UI and component-based structure)
- **Logic & AI Layer** (Implemented in Code)
- **Icons & UI Library**: Lucide-React + TailwindCSS (for modern design & responsiveness)

❖Methodology & Implementation Process System Workflow

- **Train Data Initialization**: Randomized generation of 25 train profiles with attributes (fitness, mileage, cleaning, etc.)
- **AI Scoring Engine Execution**
  → Each train's data is processed to calculate a Priority Score (0–100)
  → Status auto-classified as Ready / Standby / Maintenance
- **Alert Generation**
  → Detects expiring certificates, high job card count, and low ready count.
- **Dynamic Dashboard Display**
  → Real-time visualization of fleet readiness, statistics, and alerts.

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FEASIBILITY AND VIABILITY

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Technically Feasible: Built entirely using React.js, JavaScript, and lightweight simulation can easily integrate with real KMRL data APIs or maintenance systems (like IBM Maximo).

Operationally Practical: Simulates actual metro planning logic fitness validation, job card tracking, and bay optimization already aligned with real KMRL workflows

Cost Efficient: No hardware dependency; deployable on existing metro IT infrastructure (browser-based).

❖ Potential Challenges & Risks

- Data integration from multiple departments (Maintenance, Branding, Operations).
- Need for accurate and updated certificate/job card data from backend systems.
- System validation before real-time deployment in operational control rooms.
- Initial operator training to interpret AI recommendations.

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IMPACT AND BENEFITS

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- **KMRL Operations Team:** Gains a real-time AI dashboard for data-driven train induction decisions.
- **Maintenance & Depot Staff:** Reduced manual workload through automated readiness scoring and job card tracking.
- **Management & Planning Units:** Access to data insights on fleet performance, efficiency, and maintenance trends.
- **Passengers:** More reliable metro services due to optimized fleet availability and reduced delays.

❖ Benefits

- **Operational Efficiency:** Faster induction scheduling and fewer train readiness delays.
- **Transparency:** Every train's readiness and issues are clearly visible, improving accountability.
- **Scalability:** Framework easily adaptable to other metro systems or railway depots.

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RESEARCH AND REFERENCES

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- The solution was developed through independent analysis of real metro scheduling problems faced during daily operations.
- Ideas were formed by observing train induction challenges such as delay, certificate expiry, and depot management issues.
- Discussions with team members and brainstorming helped design the AI and manual comparison approach.

References

Personal understanding of metro operations and train scheduling logic.

Official KMRL Website (for understanding general workflow): https://kochimetro.org

Basic documentation from React.js and JavaScript (for implementation reference).

No external research papers or copied materials were used — the concept and design are originally developed by the project team.